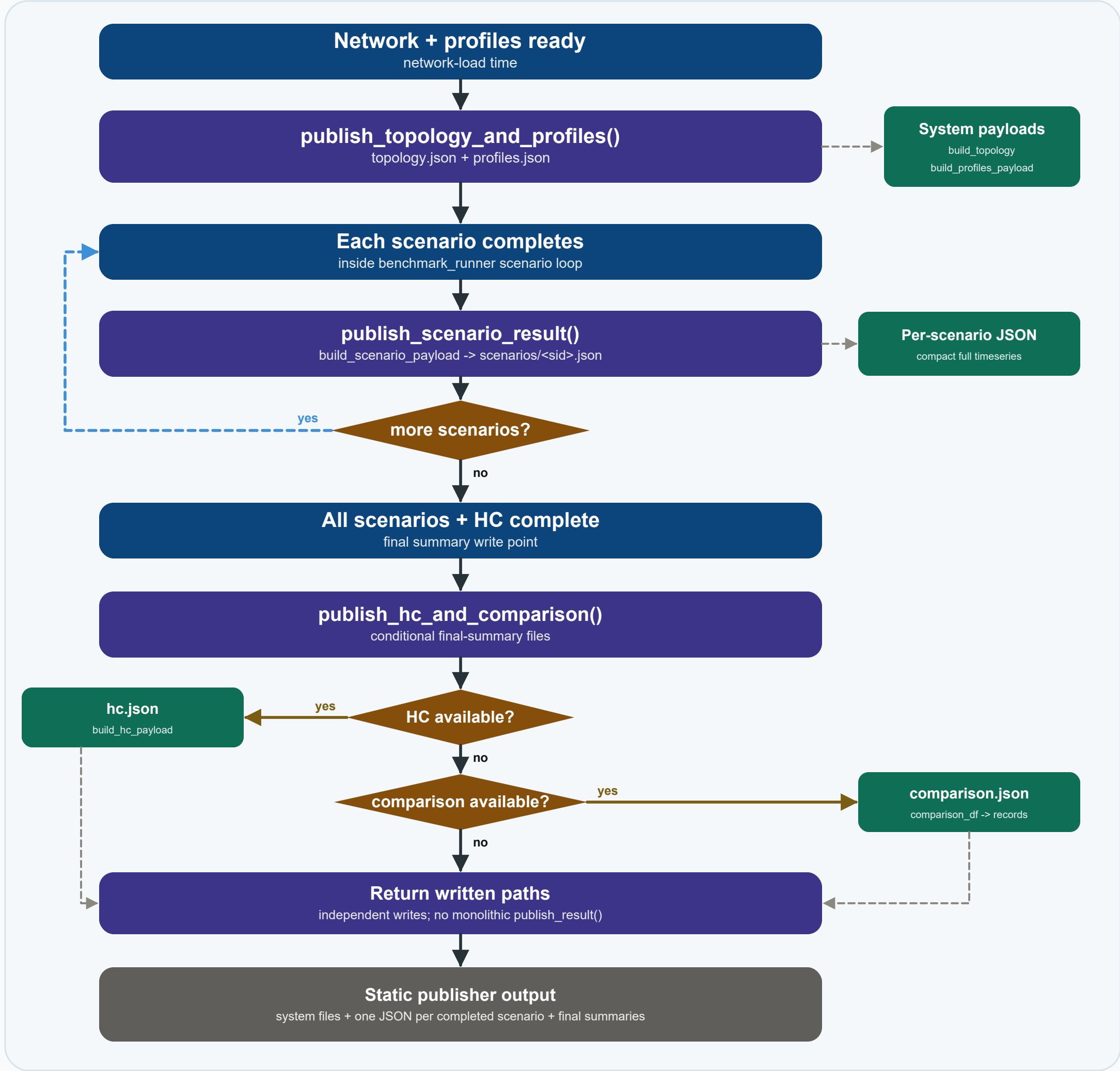




Three independent static write points

- 1. Network-load time: topology.json and profiles.json are written before scenario execution.
- 2. Scenario-loop time: each successful ScenarioResult is written immediately to scenarios/<sid>.json.
- 3. Post-run time: HC and comparison payloads are written after the scenario loop and HC analysis.

The old monolithic publish_result() flow is no longer present in the current publisher.py.

Splitting the writes improves crash resilience and avoids one large end-of-run write burst.

Payload-builder responsibilities

build_topology(): buses, lines, trafos, sgens, loads, feeder distance and voltage limits.

build_profiles_payload(): full-resolution and hourly load/PV/wind totals plus per-element series.

build_scenario_payload(): summary plus per-timestep network, violation, control and energy fields.

build_hc_payload(): baseline/Volt-Var HC summaries, gain and stored sweep-curve data.

_dump() performs JSON-safe serialization and creates parent directories as needed.

Output/file semantics

topology.json and profiles.json are system-level files.

scenarios/<sid>.json is written compactly (indent=None) because annual payloads can be very large.

publish_scenario_result(None) returns None and writes nothing.

hc.json is omitted when hc_results is empty; comparison.json is omitted when the DataFrame is empty.

The HC-stressed re-benchmark uses a separate output directory and repeats the same static write lifecycle.